BIO-233 Chapters 12-13 | Study Assistance Sheet

ONE SIDE ONLY | US Letter landscape | Print at 100% / Actual Size | Text only; draw any visuals by hand.

ORGANIZATION + CELLS

SAME: Sensory = afferent (arrives); motor = efferent (exits). **CNS:** brain + cord. **PNS:** nerves/ganglia. **ENS:** gut networks.

Somatic motor: skeletal muscle (also reflexes). **Autonomic:** smooth/cardiac muscle, glands, adipose; sympathetic/parasympathetic.

Receptors: extero = external; intero = internal; proprio = muscle/joint position.

Neuron: one cell. **Nerve/tract:** PNS/CNS axon bundle. **Ganglion/nucleus:** PNS/CNS cell bodies.

Soma: body; perikaryon = its cytoplasm. Dendrites receive; axon conducts; telodendria end in terminals. Axolemma/axoplasm = membrane/cytoplasm. Nissl = rough ER + ribosomes.

Shapes: anaxonic, no obvious axon; bipolar, 2 processes; unipolar, side soma (sensory); multipolar, many dendrites + 1 axon (motor/interneurons).

GLIA + MYELIN

CNS: astrocytes support BBB/environment, scar; ependymal line ventricles/canal, help with CSF; oligodendrocytes myelinate; microglia phagocytose.

PNS: satellite cells surround ganglion cell bodies; Schwann cells support axons and form myelin.

Oligo: segments on several axons. **Schwann:** one myelinated segment on one axon. White matter: myelinated axons; gray: cell bodies/processes.

CORD ANATOMY

31 pairs: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, 1 coccygeal. C1-7 above vertebra; C8 between C7/T1; T1 onward below.

Conus: cord ends near L1-2. Cauda equina = long roots; filum terminale = support thread.

Meninges: dura, arachnoid, pia (outer in). Epidural outside dura; CSF in subarachnoid space/central canal. Denticulate ligaments: lateral stability; filum: longitudinal.

CHANNELS: WHAT OPENS THEM?

Chemically gated (ligand gated): a chemical binds the receptor/channel and changes its opening. Example: ACh opens nicotinic cation channels at the neuromuscular junction; net inward current depolarizes the muscle.

Voltage gated: a change in membrane potential alters the channel. Axonal Na+ channels initiate the AP upstroke; K+ channels repolarize. Terminal Ca2+ channels trigger transmitter release.

Mechanically gated: physical stretch or deformation opens the channel, producing a receptor potential (touch, pressure, vibration).

Leak: open at rest; ions move down their electrochemical gradients. K+ leak strongly contributes to resting voltage. Leak channels are not one of the three gated types.

MEMBRANE POTENTIALS

Rest: about -70 mV; inside negative relative to outside. Na+ high outside; K+ high inside. **Pump:** 3 Na+ OUT, 2 K+ IN per ATP; maintains gradients.

Depolarize: less negative/more positive. **Repolarize:** return toward rest. **Hyperpolarize:** more negative than rest. Depolarization alone need not be an AP.

Graded: local, variable strength, fades with distance; depolarizing or hyperpolarizing; can sum; no AP-type refractory period.

Action potential (AP): threshold-triggered, all-or-none; regenerated; refractory period; amplitudes do not sum.

Sequence: -70 rest; about -55 mV threshold (course); voltage-gated Na+ opens, Na+ influx; about +30 peak; Na+ inactivates, K+ exits; repolarization; delayed K+ closure causes undershoot.

Absolute refractory: no new AP (Na+ channels unavailable). **Relative:** stronger stimulus needed. Refractory membrane behind favors forward travel.

Intensity: stronger sustained input usually raises AP frequency, not height. Absolute refractory period limits maximum firing rate.

CONDUCTION ALONG THE AXON

Myelinated (saltatory): myelin limits ion flow across each internode (insulated segment). Nodes of Ranvier are gaps rich in voltage-gated channels.

Node to node: local current spreads inside the axon beneath myelin, bringing the next node to threshold. Na+ channels open there and regenerate the AP.

Faster + efficient: only nodes need active AP regeneration; less membrane exchanges ions, so less pump work is needed to restore gradients.

Two speed factors: myelination reduces current leak; larger diameter lowers internal resistance. Both increase speed. Axon length alone does not; a longer route takes more travel time.

Unmyelinated (continuous): local current depolarizes the next adjacent membrane region to threshold; its voltage-gated Na+ channels open and regenerate the AP. This repeats along the axon.

No fading: passive current weakens with distance, but each new AP restores full amplitude. **Relay analogy:** each person repeats the complete message to the next; it stays intact, but every relay takes time.

Course speeds (mph): A large/myelinated, up to 268; B smaller/myelinated, ~40; C unmyelinated, ~2.

SYNAPSES + SUMMATION

Electrical: gap junctions. **Chemical:** transmitter crosses cleft. Axodendritic/axosomatic/axoaxonic = to dendrite/soma/axon.

ACh release: AP; Ca2+ channels open; influx triggers exocytosis; ACh binds nicotinic channels; depolarization. AChE splits ACh into acetate/choline; choline recycled.

Delay: Ca2+/release steps; more synapses add delay. **Fatigue:** demand exceeds replenishment.

EPSP: toward threshold. **IPSP:** inhibits (K+ out/Cl- entry). **Temporal:** repeats at one site; **spatial:** multiple sites. Graded sum: -70 + 15 = -55; -85 + 15 = -70 mV.

ROOTS, RAMI + PLEXUSES

DAVE: dorsal afferent; ventral efferent. Dorsal root ganglion (DRG): sensory cell bodies. Spinal nerves and anterior/posterior rami are **mixed**.

Endo/peri/epineurium: individual axon/fascicle/whole nerve. Posterior rami: back; anterior: body wall/limbs, form plexuses.

White rami: myelinated preganglionic sympathetic fibers (T1-L2). **Gray:** unmyelinated postganglionic return. Pre/post refers to ganglion synapse.

Course plexus ranges: cervical C1-5; brachial C5-T1; lumbar T12-L4; sacral L4-S4. **Phrenic C3-5:** diaphragm.

Musculocutaneous: arm flexors. **Median/ulnar:** many forearm/hand flexors. **Radial:** arm/forearm extensors. **Femoral:** knee extension/hip flexion. **Obturator:** adductors. Superior/inferior gluteal: medius-minimus/maximus. Sciatic: most hamstrings; tibial/common fibular branches.

NEURAL CIRCUITS + REFLEXES

Divergence: one to many. **Convergence:** many to one. **Serial:** sequence. **Parallel:** simultaneous. **Reverberation:** feedback loop.

Reflex arc: receptor; sensory neuron; CNS integration; motor neuron; effector. Mono: 1 synapse, 2 neurons; poly: interneuron(s), multiple synapses.

Classify: innate/acquired; somatic/visceral; mono/poly; spinal/cranial. Innate is inborn, still modulated by brain.

Stretch: spindle senses length; muscle contracts. Intrafusal inside spindle; extrafusal produce force; gamma adjusts spindle sensitivity.

Tendon reflex: Golgi tendon organ senses tension; inhibits the same muscle via interneurons.

Withdrawal: flexors on, extensors off (reciprocal inhibition). **Crossed extensor:** opposite leg extensors on, flexors off. Ipsi = same side; contra = opposite.

Brain: facilitates/inhibits reflexes; reinforcement enhances. Adult plantar: toes curl. Babinski: big toe extends, others fan (normal in infants; adult suggests corticospinal dysfunction).

SPACE FOR OPTIONAL HAND-DRAWN VISUALS

Baseline: supplied course sheet (textbook pp. 389-464, handouts and transcripts). Course-specific values retained; see review-deck cautions.
Channel/conduction check: OpenStax Anatomy & Physiology 2e, section 12.4. Revised November 14, 2026.